

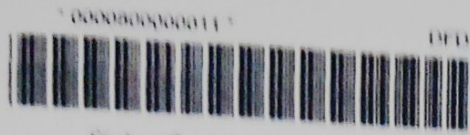
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18 $f(x) = 2x + 5$

$$f(x)f(x) = g(x) = ax^2 + bx + c$$

Find the value of a , the value of b and the value of c .

$$(2x + 5)(2x + 5) = 2(2x + 5) + 5$$

$$2x^2 + 20x + 25 = 4x + 10 + 5$$

$$2x^2 + 20x + 25 = 4x + 15$$

$$2x^2 + 16x + 10$$

$$a = \dots\dots\dots 2$$

$$b = \dots\dots\dots 16$$

$$c = \dots\dots\dots 10$$

[4]

19 Solve.

$$\left(\frac{1}{3}\right)^x = 9^{x+4}$$

$$\frac{1}{3^x} = (3^{-2})^{(x+4)}$$

$$\frac{1}{3^x} = 3^{-2x-8}$$

$$\frac{1}{3^x} = \frac{1}{3^{2x+8}}$$

$$9^x = 9^x + 8$$

$$-8 = 2x = x$$

$$-8 = x$$

$$x = \dots\dots\dots -8 \dots\dots\dots [3]$$